

BridgeTableYear

Purpose: Creates a consistent year table from 1996 to 2025 to link with all time-based fact tables.

- Ensures uniform year values
- Supports clean **one-to-many** relationships
- Improves filtering, DAX accuracy, and report performance

Code = let

 StartYear = 1996,

 EndYear = 2025,

 YearList = List.Generate(

 () => StartYear,

 each _ <= EndYear,

 each _ + 1

),

 YearTable = Table.FromList(YearList, Splitter.SplitByNothing(), {"Year"})

in

 YearTable

2. QuarterTable

Purpose: Generates combinations of Year and Quarter (e.g., 2018Q1) from 2018 to 2025.

- Enables seasonal analysis
- Links cleanly to quarterly GDP
- Avoids redundant year/quarter logic in reports

Code = let

```

StartYear = 2018,
EndYear = 2025,
Quarters = { "Q1", "Q2", "Q3", "Q4" },
YearQuarterList = List.Combine(
    List.Transform(
        {StartYear..EndYear},
        each List.Transform(Quarters, (q) => Text.From(_) & q)
    )
),
YearQuarterTable = Table.FromList(YearQuarterList, Splitter.SplitByNothing(),
{"YearQuarter"})
in
    YearQuarterTable

```

3. DimSector

Purpose: Combines and cleans Sector columns from multiple fact tables to create a unified dimension.

- Resolves **many-to-many** issues with a **one-to-many** model
- Standardizes naming (trimming, casing)
- Enables consistent filtering and accurate DAX

Code = let

```
// Select and rename Sector columns for consistency
```

```

GDP_Annual =
Table.RenameColumns(Table.SelectColumns("#Panama's_Annual_GDP__2018_2023",
{"Sector"}), {"Sector", "Sector"}),

```

```

GDP_Quarterly =
Table.RenameColumns(Table.SelectColumns("#Panama's_Quarterly_GDP__2018_2023",
{"Sector"}), {"Sector", "Sector"}),

```

```

    FDI = Table.RenameColumns(Table.SelectColumns("#FDI__by_Economic_Activity",
{"Sector"}), {"Sector", "Sector"}),

    GVA =
Table.RenameColumns(Table.SelectColumns("#Gross_Value_Added__GVA__by_Economic
_Sector", {"Sector"}), {"Sector", "Sector"}),

    GDP_Historical = Table.RenameColumns(Table.SelectColumns("#Panama's GDP by
Economic Activities", {"Sector"}), {"Sector", "Sector"}),

// Combine all sectors

AllSectors = Table.Combine({GDP_Annual, GDP_Quarterly, FDI, GVA, GDP_Historical}),

// Clean: Trim whitespace, remove nulls/empty, and lowercase for standardization

CleanedSectors = Table.SelectRows(AllSectors, each [Sector] <> null and
Text.Trim([Sector]) <> ""),

TrimmedSectors = Table.TransformColumns(CleanedSectors, {"Sector", Text.Trim}),

LowerCased = Table.TransformColumns(TrimmedSectors, {"Sector", Text.Lower}),

// Remove duplicates

UniqueSectors = Table.Distinct(LowerCased),

// Capitalize sector names again (optional)

Capitalized = Table.TransformColumns(UniqueSectors, {"Sector", Text.Proper}),

// Add SectorKey

AddKey = Table.AddIndexColumn(Capitalized, "SectorKey", 1, 1, Int64.Type)
in

AddKey

```

DAX Calculations

1. FDI_Economic_Activity

DAX

Calculate Total FDI

1. 1Total FDI = SUM('FDI_Economic_Activities'[FDI_Value])

Use on

CARD VISUAL.

1. Total FDI 2018–2022 =

CALCULATE(

[Total FDI],

FILTER(

ALL('BridgeTableYear'),

'BridgeTableYear'[Year] >= 2018 && 'BridgeTableYear'[Year] <= 2022

)

)

For sector comparison.

To Track the performance of FDI in panama over the years

(A) Calculating year-over-year FDI Growth Rate (%).

FDI Growth Rate (%) =

VAR CurrentYear = SELECTEDVALUE(BridgeTableYear[Year])

VAR PrevYear = CurrentYear - 1

VAR IsInRange = CurrentYear >= 2017 && CurrentYear <= 2023

VAR PrevFDI =

CALCULATE(

```

SUM(.FDI_By_Economic_Activity[FDI_Value]),
FILTER(
    ALL(BridgeTableYear),
    BridgeTableYear[Year] = PrevYear
)
)

```

```

VAR CurrFDI =
CALCULATE(
    SUM(.FDI_By_Economic_Activity[FDI_Value]),
    FILTER(
        ALL(BridgeTableYear),
        BridgeTableYear[Year] = CurrentYear
    )
)

```

```

RETURN
IF(
    IsInRange && NOT ISBLANK(PrevFDI) && PrevFDI > 0,
    DIVIDE(CurrFDI - PrevFDI, PrevFDI) * 100,
    BLANK()
)

```

Visual.

Line chart

To Determin the Compound Annual growth rate.

FDI 1CAGR (%) =

VAR StartYear = 2017

VAR EndYear = 2023

VAR NumYears = EndYear - StartYear

VAR StartValue =

```
CALCULATE(  
    [Total FDI],  
    FILTER(  
        ALL('BridgeTableYear'),  
        'BridgeTableYear'[Year] = StartYear  
    )  
)
```

VAR EndValue =

```
CALCULATE(  
    [Total FDI],  
    FILTER(  
        ALL('BridgeTableYear'),  
        'BridgeTableYear'[Year] = EndYear  
    )  
)
```

VAR CAGR =

```
IF(  
    NOT ISBLANK(StartValue) && StartValue > 0 && NOT ISBLANK(EndValue),  
    (EndValue / StartValue) ^ (1 / NumYears) - 1,
```

```
BLANK()  
)
```

RETURN

ROUND(CAGR * 100, 1)

Visual

Table matrix

2. FDI_by_top_10_Countres (Top partner countries)

Top Contributing Countries to Panama's GDP

DAX:

Total FDI Amount =

SUM (FDI_by_top_10_Countries [FDI_Value])

CARD VISUAL.

To track growth performance of FDI Year-over-Year Change (%).

FDI YoY Change % =

VAR CurrentYear = SELECTEDVALUE(BridgeTableYear[Year])

VAR PrevYear = CurrentYear - 1

VAR CurrentFDI =

CALCULATE(

[Total FDI Amount],

BridgeTableYear[Year] = CurrentYear

)

VAR PrevFDI =

CALCULATE(

[Total FDI Amount],

BridgeTableYear[Year] = PrevYear

)

RETURN

IF(

CurrentYear >= 2017 && CurrentYear <= 2023,

DIVIDE(CurrentFDI - PrevFDI, PrevFDI, 0),

BLANK()

)

Visual.

LINE CHART OR METRIX.

3. Gross_Value_Added_GVA_by_Economic_Sector.

Dax:

Calculating the total gross value added

Total GVA = SUM(Gross_Value_Added__GVA__by_Economic_Sector[GVA_Value])

Visualize with card visual.

To calculate the growth%

measure for calculating **Growth % from 2018 to 2022**

Growth % 2018–2022 =

VAR GVA_2018 =

CALCULATE(


```

[Total GVA],
FILTER(
    ALL('BridgeTableYear'),
    'BridgeTableYear'[Year] = 2018
)
)
VAR GVA_2022 =
CALCULATE(
    [Total GVA],
    FILTER(
        ALL('BridgeTableYear'),
        'BridgeTableYear'[Year] = 2022
    )
)
RETURN
IF(
    GVA_2018 > 0,
    DIVIDE(GVA_2022 - GVA_2018, GVA_2018),
    BLANK()
)

```

Calculating CAGR (Compound Annual Growth Rate) for 2018–2022

CAGR % 2018–2022 =

VAR StartYear = 2018

VAR EndYear = 2022

VAR NumPeriods = EndYear - StartYear

VAR GVA_2018 =

```
CALCULATE(  
    [Total GVA],  
    FILTER(  
        ALL('BridgeTableYear'),  
        'BridgeTableYear'[Year] = StartYear  
    )  
)
```

VAR GVA_2022 =

```
CALCULATE(  
    [Total GVA],  
    FILTER(  
        ALL('BridgeTableYear'),  
        'BridgeTableYear'[Year] = EndYear  
    )  
)
```

RETURN

```
IF(  
    NOT(ISBLANK(GVA_2018)) && GVA_2018 > 0,  
    POWER(GVA_2022 / GVA_2018, 1 / NumPeriods) - 1,  
    BLANK()  
)
```

Visual.

line charts

To Calculate Sector contribution by CAGR%

GVA CAGR % 2018–2022 =

VAR StartYear = 2018

VAR EndYear = 2022

VAR NumPeriods = EndYear - StartYear

VAR GVA_2018 =

```
CALCULATE(  
    [Total GVA],  
    FILTER(  
        ALL('BridgeTableYear'),  
        'BridgeTableYear'[Yearly] = StartYear  
    )  
)
```

VAR GVA_2022 =

```
CALCULATE(  
    [Total GVA],  
    FILTER(  
        ALL('BridgeTableYear'),  
        'BridgeTableYear'[Yearly] = EndYear  
    )  
)
```

RETURN

```

IF(
    NOT ISBLANK(GVA_2018) && GVA_2018 > 0,
    (POWER(GVA_2022 / GVA_2018, 1 / NumPeriods) - 1),
    BLANK()
)

```

Visual Bar Chart

4. Panama's GDP by Economic Activity (1996–2022)

Dax:

Calculating Total GDP = SUM('GDP_Economic_Activity'[GDP_Value])

Tracking sectoral performance from 1996-2022

CAGR % 1996–2022 =

VAR StartYear = 1996

VAR EndYear = 2022

VAR NumPeriods = EndYear - StartYear

VAR GDP_1996 =

```

CALCULATE(
    [Total Sector GDP],
    FILTER(
        ALL('BridgeTableYear'),
        'BridgeTableYear'[Yearly] = StartYear
    )
)

```

VAR GDP_2022 =

```

CALCULATE(
    [Total Sector GDP],
    FILTER(
        ALL('BridgeTableYear'),
        'BridgeTableYear'[Yearly] = EndYear
    )
)

```

```

RETURN
IF(
    NOT ISBLANK(GDP_1996) && GDP_1996 > 0,
    POWER(GDP_2022 / GDP_1996, 1 / NumPeriods) - 1,
    BLANK()
)

```

Visual.

Stacked bar chart.

Tracking sectoral performance from 2018-2022

1CAGR % 2018–2022 =

VAR StartYear = 2018

VAR EndYear = 2022

VAR NumPeriods = EndYear - StartYear

VAR GDP_2018 =

```

CALCULATE(
    [Total Sector GDP],

```

```

    FILTER(
        ALL('BridgeTableYear'),
        'BridgeTableYear'[Yearly] = StartYear
    )
)

VAR GDP_2022 =
    CALCULATE(
        [Total Sector GDP],
        FILTER(
            ALL('BridgeTableYear'),
            'BridgeTableYear'[Yearly] = EndYear
        )
    )

RETURN
IF(
    NOT ISBLANK(GDP_2018) && GDP_2018 > 0,
    (POWER(GDP_2022 / GDP_2018, 1 / NumPeriods) - 1) * 100,
    BLANK()
)

```

5. Panama's_Annual_GDP__2018_2023:

Dax:

Calculating Total Annual GDP:

Total GDP 2018-2023 =

```
CALCULATE(
    SUM('Panama's_Annual_GDP__2018_2023'[GDP_Annual_Value]),
    FILTER(
        'Panama's_Annual_GDP__2018_2023',
        'Panama's_Annual_GDP__2018_2023'[Year] >= 2018 &&
        'Panama's_Annual_GDP__2018_2023'[Year] <= 2023
    )
)
```

calculate the GDP Growth Rate (%).

GDP Growth Rate (%)2 =

VAR CurrentYear = SELECTEDVALUE(BridgeTableYear[Year])

VAR CurrentGDP =

```
CALCULATE(
    SUM('Panama's_Annual_GDP__2018_2023'[GDP_Annual_Value]),
    FILTER(
        ALL('Panama's_Annual_GDP__2018_2023'),
        'Panama's_Annual_GDP__2018_2023'[Year] = CurrentYear
    )
)
```

VAR PreviousGDP =

```
CALCULATE(
    SUM('Panama's_Annual_GDP__2018_2023'[GDP_Annual_Value]),
    FILTER(
```

```

    ALL('Panama's_Annual_GDP__2018_2023'),
    'Panama's_Annual_GDP__2018_2023'[Year] = CurrentYear - 1
)
)

RETURN

IF(
    NOT(ISBLANK(CurrentGDP)) && NOT(ISBLANK(PreviousGDP)) && PreviousGDP <> 0,
    DIVIDE(CurrentGDP - PreviousGDP, PreviousGDP),
    BLANK()
)

Visual.

line chart

```

6 Panama Provincial_GDP_2018–2023:

Dax:

Calculating Top panama's Provinces by Total GDP.

Total_GDP_By_province = SUM('Provincial_GDP'[GDP_Value]).

Province GDP by Year over Year Growth.

GDP_YoY =

VAR CurrentYear = MAX('BridgeTableYear'[Year])

VAR PrevYear = CurrentYear - 1

VAR CurrGDP =

CALCULATE(
[Total_GDP],


```

    'BridgeTableYear'[Year] = CurrentYear
)
VAR PrevGDP =
    CALCULATE(
        [Total_GDP],
        'BridgeTableYear'[Year] = PrevYear
    )
RETURN
    DIVIDE(CurrGDP - PrevGDP, PrevGDP)

```

Visual.

Map of panama

7. Panama's_Quarterly_GDP_2018–2023.

DAX:

Calculating the Average GDP By Quarter

```

Avg_GDP_By_Quarter =
    AVERAGEX(
        VALUES(BridgeTableQuarter[Quarters]),
        CALCULATE(SUM('Panama's_Quarterly_GDP__2018_2023'[Quarter_GDP_Value]))
    )

```

Visual.

Line chart.

Tracking Quarter growth

Quarterly_GDP_Growth_% =

```
VAR CurrentQ = SELECTEDVALUE(BridgeTableQuarter[Quarters])
```

```
VAR IsInRange =
```

```
    CurrentQ >= "2018Q1" && CurrentQ <= "2024Q3"
```

```
VAR PrevQ =
```

```
    CALCULATE(
```

```
        MAX(BridgeTableQuarter[Quarters]),
```

```
        FILTER(
```

```
            ALL(BridgeTableQuarter),
```

```
            BridgeTableQuarter[Quarters] < CurrentQ
```

```
        )
```

```
    )
```

```
VAR CurrGDP =
```

```
    CALCULATE(
```

```
        SUM('Panama's_Quarterly_GDP__2018_2023'[Quarter_GDP_Value]),
```

```
        TREATAS({CurrentQ}, 'BridgeTableQuarter'[Quarters])
```

```
    )
```

```
VAR PrevGDP =
```

```
    CALCULATE(
```

```
        SUM('Panama's_Quarterly_GDP__2018_2023'[Quarter_GDP_Value]),
```

```
        TREATAS({PrevQ}, 'BridgeTableQuarter'[Quarters])
```

```
    )
```

```
RETURN
```

```
IF(  
    IsInRange && NOT ISBLANK(PrevGDP),  
    DIVIDE(CurrGDP - PrevGDP, PrevGDP),  
    BLANK()  
)
```

visual

Table matrix